

Finals Task 2. Working and Navigating through Power BI

Instruction: This task will allow you to navigate and put together the Power BI workflow from Data Preparation, Modeling, Analysis and Visualization.

Part 1. Data Acquisition. Download the **flights_data.csv** and **airports.csv** for the dataset that you need to work on, your task is to clean and prepare the data ready for modeling. The dataset is currently in a form of flat csv file; you are to check and correct necessary data cleaning task.

1. Make sure all columns are properly labeled
2. You may replace all blank values under cancellation reason to "Security"
3. Remove unnecessary columns specially those that contains blank entries
4. Split necessary dim tables for the airline, cancellation_reason, airports
5. Add from the data source the airports.csv file

Double check the columns in your tables (You should have 1 fact table and 3 dim tables)

Part 2. Data Model – Design the data model – using Star Schema

Part 3. Do the necessary Measures and Calculated Columns use DAX and organized your measures into folders (by going to the Model view-Click Data) then under properties – add a folder named measures and put all measures in the folder

Here are the measures:

Calculated Columns: (Add Column from the Data view)

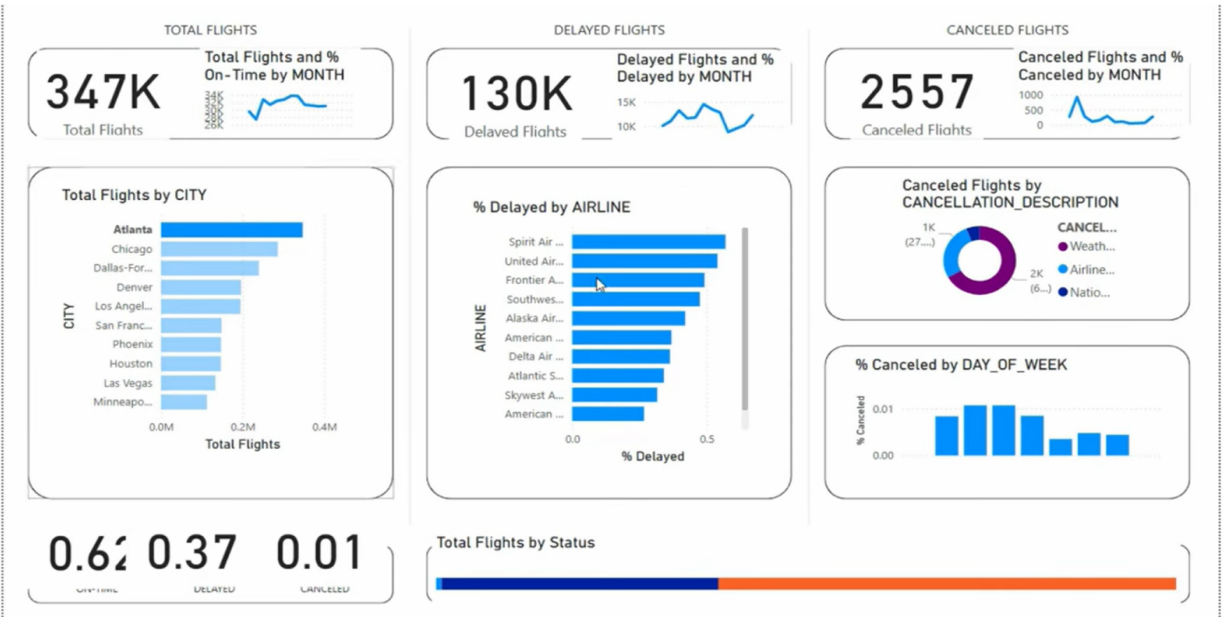
STATUS = IF(flights[CANCELLED]=1,"Canceled",
IF(flights[DEPARTURE_DELAY]>0,"Delayed", "On-Time"))

- **Total Flights** = COUNTROWS(flights)
- **% Canceled** = DIVIDE([Canceled Flights],[Total Flights],"-")
- **% Delayed** = DIVIDE([Delayed Flights],[Total Flights],"-")
- **% On-Time** = DIVIDE([On-Time Flights],[Total Flights],"-")
- **Total Canceled Flights** = CALCULATE([Total Flights], flights[STATUS]="Cancelled")
- **Total Delayed Flights** = CALCULATE([Total Flights], flights[STATUS]="Delayed")
- **Total On-Time Flights** = CALCULATE([Total Flights], flights[STATUS]="On-Time")

Part 4. Do the necessary visuals for all the measures computed



Draft



Final Design – Explore the formatting tool of Power Bi visuals

